

Intero-Expressive Robots: Externalizing Robot Internal Signals for Exploring Theory of Mind Attribution to Robots

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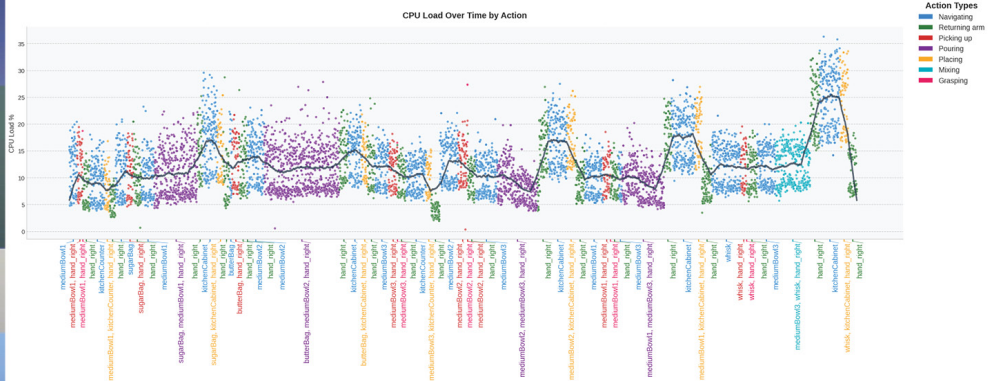


Figure 1: Left: The virtual kitchen environment with a robot performing a cooking task. Right: agentic load, describing the fluctuations in the robot’s CPU load during the task. Different actions are indicated by color as described in the legend. Labels on the X time axis represent the variable for each of the actions, e.g. Navigating to (KitchenCabinet).

Abstract

We present the concept of Intero-Expressive Robots, proposing external representation of internal signals to make hidden states perceivable during interaction. In humans, external indications of interoceptive signals such as respiration or cardiovascular function support attribution of internal states to others (Theory of Mind) by

providing information that cannot be inferred from behavior alone. Drawing on interoception as a design analogy, we discuss how exposing robotic variables—such as computational load or energy use—may support analogous forms of state attribution. We position intero-expression as complementary to robot transparency and legibility: externalizing internal dynamics that shape behavior and constrain performance. We describe preliminary feasibility work in a virtual environment, where a robot executes a task while a CPU-derived internal signal is logged. Identical actions produced different loads depending on task context, patterns not predictable from observation alone. This suggests that intero-expressive representations can be interaction-relevant cues.



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CCS Concepts

• **Human-centered computing** → **Interaction techniques; Interaction design; Interactive systems and tools**; • **Computer systems organization** → **Robotic control**.

Keywords

Theory of Mind, Embodied cognition, Interoception, Robotic internal states, Social robotics

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1 Introduction and related work

Social interaction depends on the ability to attribute mental states to others, a capacity called Theory of Mind (ToM) [13, 17]. ToM supports prediction, coordination, and communication. It was originally defined as the capacity to attribute mental states such as beliefs and desires to others, and to recognize that these may differ from one's own [23]. In Human-Robot Interaction (HRI), people may also interpret robot behavior through ToM-like inference: people ascribe mental features to robotic agents [4] often discussed in terms of anthropomorphism and mind perception [12, 29], and interactions with robots activate brain regions linked to social perception [30]. Such inference can arise even when the counterpart has no genuine inner states. Humans are already inclined to interpret machines socially [1, 32], as also suggested by the Computers as Social Actors (CASA) framework, which finds that people apply social heuristics to artificial systems by default [19]. In addition, reviews show that people readily ascribe beliefs, desires, and intentions to robots under a variety of conditions [27]. Recent work on Theory-of-Mind reasoning by robots has also applied such models in robot-robot interaction settings [5]. As robots move into more social roles, users need to form workable internal models of what the robot is doing. This is easier when relevant internal processes are made legible, rather than kept as hidden computation [15].

In many interaction settings, collaboration depends on whether users can infer what the robot is “doing” and what it is likely to do next, alongside factors such as appearance and presentation [1, 32]. This fits with findings from human-AI collaboration research, where ToM-related skills predict joint performance [24]. Users who can reason about a system's perspective, confidence, timing, or limitations collaborate better. One implication for HRI is that externalizing aspects of a robot's internal processes may support better anticipation and coordination, whether or not users interpret this socially. Several researchers have proposed that HRI could benefit from making internal processes more transparent and interpretable [11, 15]. In considering how internal processes might be conveyed, we can draw inspiration from how internal states become perceptible in humans.

Interoceptive (internal) physiological signals are tied to awareness of internal states [9]. In humans, these signals often have visible external correlates: breathing patterns can be seen and heard, and thermoception shows up as goosebumps, sweating, or shifts in skin tone. Cardioception, too, surfaces in facial expressions [2]. These recognizable signals form part of how we develop ToM. High heart rate, or blushing, can signal excitement. Breath rate tracks stress or calmness. These cues matter because internal states are not fully predictable from behavior: someone walking slowly might be tired, careful, or deep in thought. The visible physiological signal disambiguates. We read these cues because we know how they map onto our own internal states [21]. Here, we use interoception as a design inspiration for how internal dynamics can become perceptible during interaction, rather than as a claim of equivalence between physiology and computation.

We propose that external representations of a robot's internal signals could serve a similar interaction function by improving transparency and legibility during collaboration. Robots lack mental states, but they have hidden inner processes that shape their behavior. We suggest that a robot's internal variables, such as temperature, power consumption, and CPU load, can be treated as internal state information that fluctuates during task execution and constrains performance. High current draw might indicate a physically demanding task. Elevated CPU load could indicate computational complexity. Low battery might parallel exhaustion. Human interoception covers a broad range of sensations: pain, digestion, and more. Robotic interoceptive information could similarly include signals such as component temperatures, wear indicators, and communication delays [18]. By giving these signals an external representation, we aim to let humans perceive internal dynamics that would otherwise be invisible, helping them anticipate behavior and calibrate expectations.

The relationship between internal state visibility and social interpretation is often discussed in terms of embodiment. Human experience is deeply embodied; cognitive and mental states are linked to bodily representation [14]. Robots have a different form of embodiment, spanning physical components and computational processes. A robot that appears to have no internal constraints or self-monitoring may be perceived as opaque or “zombie-like” [8]. Prior HRI work has explored physiology-like signals in robots [3, 31]. Our aim, on the other hand, is not to induce mental state attribution through mimicry. Instead, we suggest conveying information about how the robot is functioning, using signals that arise from its own operation, and that could be learned through repeated exposure.

As a first step in exploring this concept, we created a VR environment with a robot action simulator and recorded an internal signal of the robot (*agentic load*, defined below, as a representation of CPU load) while the robot performed a scripted cooking task. This feasibility study is intended to provide intuition for the dynamics of such signals and to motivate possible external representations. In this paper, we describe the concept, the implementation, and the observed signal patterns. A key finding is that internal dynamics were not predictable from behavior: identical actions produced different loads depending on task context. This suggests that such signals carry information that observation alone cannot provide, paralleling why physiological cues matter in human interaction.

We outline future work to test whether people can interpret and use these signals.

2 Feasibility Study

Our goal in this feasibility test was to check a premise: do internal signals carry information not already visible from behavior? In order to have a fully controlled environment, we chose to do this in a VR setup that includes an advanced robot action simulator (see Figure 1). We recorded an interoceptive signal during a robot’s operation and plotted the signal, giving an indication of how the representation of such signals could expose meaningful information on the robot’s internal state.

2.1 Technical Implementation

To produce a representative log of a robotic agent’s internal signal during an interaction, we used an existing VR kitchen environment developed for collaborative task experiments in a simulated kitchen [20, 25]. The VR kitchen, developed using the pyBullet physics engine and visualized using the Unity game engine, allows users to interact with a virtual robot that performs cooking tasks autonomously based on high-level commands. The physics engine allows modeling of rigid body mechanics, and for more complex phenomena involved in cooking (e.g. an object being cut or baked), object behavior is approximated via “custom dynamics” which are triggered by specific motions of tools relative to affected objects. While the robot can be given commands by a human user in the environment, for our purposes, we used a script to provide a sequence of commands to the robot. The commands are a series of steps to perform a recipe. These steps are fairly high-level, but require the objects involved to be specified. E.g., while in a recipe book one may see instructions such as “cut an onion”, the robot would receive a command similar to “cut onion_1 using knife_2 and deposit the results in bowl_3”.

In our scenario, the robot was given sequential high-level instructions following an almond cookie recipe which was included in the environment [20]. While the robot in the underlying simulation was receiving the high-level commands for the recipe steps (e.g. “Fetch bowl_1 to the table”, “mix the flour and the sugar located in bowl_1”), a log was collected of the *agentic load*. This load is the ratio of the time spent performing the computations necessary to control the robot – checking relevant world state variables, deciding what to do, computing values for actuators – to the time spent computing a state update for the simulated world (which includes the robot control, the physics engine calculations, and the custom dynamics). The agentic load at various times is associated with the lower-level actions the robot was completing (e.g. “Navigating to bowl”, “Moving arms to correct location to pick up the bowl”, “Grasping the bowl”). Our simulation included a total of seven action types: Navigating, Returning arm (to default position), Picking up item, Pouring, Placing item, Mixing, and Grasping. These can be applied to different variables (e.g. “Picking up bowl_1”, or “Moving left arm to default position”).

The simulated world state update was performed in a loop, where each iteration corresponds to a simulation timestep. For each timestep, physics engine and custom dynamics are invoked to compute updates to variables. Thus, while the computation is done

together, one can still separate the time taken by the various sub-systems and in particular distinguish intervals of time associated with computing robot updates. This allows the agentic load to be considered as an interoceptive signal of the simulated robot.

2.2 Results

The logged data clearly shows that agentic load dynamically fluctuates with task demands, while showing a consistent pattern for each action as seen in Figure 1. This indicates the signals reflect aspects of the robot’s internal workload. It is important to note that these patterns were not always intuitive. Not only did the same actions performed on different values lead to different agentic loads (e.g. picking up a bowl or butter), identical low-level actions could also produce different computational loads depending on contextual factors. For example, as can be seen in Figure 1, navigating from the kitchen counter to the shelf to fetch a bowl in the beginning of the recipe requires low agentic load. However, later in the recipe, when navigating between the same two locations again to fetch a whisk, a much higher agentic load was recorded. We believe this is due to the robot being programmed to keep track of relevant variables from the state of the world, and this set of relevant variables changes throughout task execution. In particular, as a task progresses, the robot checks that previously obtained results remain valid for as long as they are needed (e.g., that a plate brought to the table has remained on the table). A further factor that was recognized in the detailed logs as influencing agentic load was the computation of arm motion, which used an iterative inverse kinematics solver. For a particular low-level action, in a particular context, most of the variability in agentic load was due to variability of time taken by inverse kinematics and this variability would be fairly rhythmic. At somewhat regular intervals, this solver would need many iterations, possibly to correct for errors accumulated in previous timesteps in which it performed only a few iterations.

These findings confirm the premise: internal signals carry information that observing behavior alone cannot provide. This provides a motivation for their externalization as meaningful interaction cues.

3 Discussion

Intero-expressive cues offer a way to make robot internal dynamics perceptible as part of interaction, rather than as post hoc explanations or symbolic status indicators. As discussed in the introduction, humans already do attribute inner states to robots despite the robots’ lack of an actual mental state. We believe intero-expression could support forming mental models that are more meaningful and intuitively help users gain a better understanding of the robot’s actual functioning.

Our findings suggest these dynamics are worth externalizing: internal load varied with task context in ways not predictable from the robot’s visible actions. Signals such as joint temperature, computational load, or battery strain becoming externally visible may help users gain insight into how a robot is currently operating, and may support anticipation of timing, reliability, or limitations. In this sense, intero-expression can be understood as a form of situated transparency, where otherwise hidden internal processes become



Figure 2: Illustrations of possible intero-expressive visualizations. From left to right: Dimming head light; Sonification; Dimming chest light

part of the interactional context. This aligns with prior work showing that making aspects of robot architecture and operation legible can support coordination in HRI [15].

It is important to note that displaying indications of robotic internal signals is not new; for example, battery level indicators are common. However, such signals are typically framed as diagnostic status information rather than as part of the interaction itself. By situating these signals in the broader context of intero-expression—making internal dynamics continuously perceivable and interpretable during interaction—we reframe their role as supporting intuitive understanding of a robot’s inner state. This perspective helps motivate new design directions and guidelines for integrating internal signals as embodied, interaction-relevant cues rather than symbolic indicators.

The approach explored here relies on signals that arise from the robot’s own physical or computational substrate, rather than imitating human physiology or emotional expression. This resonates with work on robotic otherness [6, 7, 26], which argues for designing robots as partners with their own modes of embodiment rather than as approximations of human behavior. From this perspective, collaboration emerges from learning how a particular system works, not from making it appear human-like or mentally transparent in a literal sense. This also aligns with recent explorations surrounding “robotic phenomenology” [16, 22].

Rather than translating internal states into speech, icons, or affective displays, intero-expressive signals may be learned through repeated exposure, similar to how people become attuned to the idiosyncratic cues of tools, animals, or collaborators with unfamiliar sensory styles.

For example, users might come to recognize when increased computation slows a movement, when power use rises during precision work, or when planning exhibits characteristic temporal patterns. In such cases, coordination is supported by reading ongoing dynamics, not by explicit explanation.

Figure 2 illustrates several candidate representations of intero-expressive signals, intended as design sketches rather than evaluated interfaces. These examples explore how internal dynamics

might be localized on the robot’s body and mapped across modalities. For instance, changes in light intensity on the head or chest could reflect increasing computational or energetic load, while sonification could convey temporal fluctuations or variability in internal processing. Such representations differ in how immediately interpretable they are, how they scale with signal intensity, and how they compete with other perceptual demands during interaction. Importantly, these sketches are not meant to prescribe specific mappings, but to highlight the design space and tradeoffs involved in making internal dynamics perceptible without resorting to symbolic dashboards or verbal explanation.

This work has clear limitations. First, as an exploratory feasibility study, it does not include an evaluation with participants, and therefore does not test whether people can interpret intero-expressive signals, whether such interpretation is intuitive or learned, or how it affects interaction. Second, we investigated only a single signal type, agentic load, derived from CPU-related computation. Other internal variables such as component temperatures, current draw, actuator strain, communication latency, or wear remain unexplored. Third, the study was conducted in a VR environment, which allowed precise control and logging but limits claims about embodiment and real-world interaction with physical robots.

Future work should therefore focus on evaluating whether and how humans can make use of intero-expressive signals during interaction. Controlled studies could examine whether such signals support anticipation, reduce coordination errors, or improve timing in collaborative tasks, and how these effects compare to more conventional transparency mechanisms. An important design question concerns representation: intero-expressive signals could be localized to different parts of the robot, mapped across modalities such as light, sound, vibration or motion, and presented at different temporal resolutions. As with physiological signals in humans, internal dynamics could be represented directly or through processed summaries, such as trends or variability [10].

A further direction concerns reciprocity. Rather than treating intero-expression as a one-way display, future systems could adapt their behavior based on sensed human state, such as heart rate or skin conductance, adjusting timing or strategy accordingly, as explored in recent HRI work [28]. In such cases, intero-expressive signals become part of a feedback loop in which both human and robot adapt to each other’s changing internal conditions. This framing does not require assuming shared mental states or intentions, but instead focuses on mutual adjustment through observable dynamics.

4 Conclusion

This paper introduced the concept of intero-expressive cues as a means of externalizing robotic internal states to support ToM-like inference in HRI. By grounding the idea in interoceptive theory and embodied cognition, and illustrating its feasibility through preliminary simulation work, we aim to open a research direction connecting robotic self-representation, transparency, and social cognition. Intero-expressive cues may ultimately support more intuitive, trustworthy, and empathetic human-robot relationships.

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